

Chris Dollo

Halethorpe, MD | dollochrisdavid@gmail.com | linkedin | github | Portfolio

EDUCATION

University of Maryland, College Park

M.S. Computer Science

College Park, MD

Aug 2026 – May 2027

University of Maryland, Baltimore County

B.S. Computer Science

Catonsville, MD

Jan 2022 – Dec 2025

EXPERIENCE

Research Assistant

Jan 2024 – May 2026

UMBC Sensorimotor Control Lab (Dr. Ramana Vinjamuri)

Catonsville, MD

- Led an **in-depth, independent research project** on unique real-world motion datasets, architecting a real-time **Python** analysis pipeline (**OpenCV**, **MediaPipe**) over **50+ gesture classes** to extract quantitative motion primitives.
- Developed **Gaussian statistical models** of joint angular velocity to **extract latent signals** (“syllables of motion”), achieving **95.78%** reconstruction accuracy for natural grasps and **92.99%** for mudras – **outperforming published baseline methodologies** by up to **10.48%**.
- Designed and empirically benchmarked **ML classifiers** (**SVM**, **LSTM**, **Random Forest**) on **multi-channel EMG sensor datasets**; ran **ablation studies** and **hypothesis-driven evaluation**, documenting and communicating findings to collaborators.
- Publication:** C. Dollo et al. “Reconstructing Hand Gestures with Synergies from Dance Movements.” *Nature Scientific Reports*, 2025. (2nd Author, Accepted)

Data Scientist Intern

May 2026 – Aug 2026

Caterpillar

Chicago, IL

- Engineered an automated **LLM evaluation framework** in **Python** (**Pytest**, **GitHub Actions**) to **quantify AI assistant quality** after every commit, enabling rigorous, reproducible regression testing of model performance.
- Built scalable **data processing pipelines** leveraging **Whisper**, **GPT-4o Transcribe**, and multiple speech and language models to preprocess audio, generate transcripts, and support **metric-driven evaluation** of agent performance across large conversation datasets.
- Deployed Dockerized evaluation services (**Docker Compose**, **microservices**), partnering with engineers and stakeholders to deliver reproducible experimentation infrastructure for production machine learning systems.

Machine Learning Intern

May 2025 – Aug 2025

Verizon

Catonsville, MD

- Designed and deployed an end-to-end computer vision pipeline using **YOLOv8** on **14,000+ labeled images**, reaching **94% accuracy** and eliminating manual review for 3 product lines.
- Preprocessed and augmented **real-world datasets** with rotation, scaling, and contrast transforms, applying **empirical error analysis** to improve robustness and **reduce false detections by 20%**.
- Optimized **PyTorch** training and inference workflows, cutting average latency by **35%** and enabling real-time deployment on edge hardware.
- Built automated **unit and integration test suites**, validating **100%** of production inference endpoints before deployment.

PROJECTS

EcoRecyclr | PyTorch, Python, ResNet-18, Flask, Android (Kotlin), REST API

Sep 2025 – Oct 2025

- Led a **4-member team** to **1st place** at the Booz Allen Hamilton Code for Good Hackathon, delivering real-time classification of waste recyclability with on-device ML inference.
- Fine-tuned **ResNet-18** in **PyTorch** on **10,000+ images**, achieving **95% accuracy** across 5 categories within a **48-hour** window; architected background **Kotlin** services sustaining **60fps** UI responsiveness.

Hellacious | React.js, Flask, MongoDB, Firebase, AWS S3, Node.js, Tailwind CSS

Oct 2025 – Present

- Designed and shipped a full-stack social platform with **15+ active users**, covering the full **SDLC** from requirements to production – auth flows, feeds, and media uploads.
- Integrated **Firebase Auth** and **RESTful APIs** for secure session management across **3 user roles**; deployed media on **AWS S3** with lifecycle policies, reducing API response time by **40%** under peak load.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C++, Kotlin, JavaScript, TypeScript, Bash, HTML/CSS

Data & ML Libraries: NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, OpenCV, MediaPipe, Flask, React.js, Next.js, Node.js, Express.js, Tailwind CSS

Databases: PostgreSQL, MySQL, SQLite, MongoDB

Cloud & Tooling: Linux, Docker, Git, GitHub Actions, CI/CD, AWS (EC2, S3, Lambda), Firebase

Concepts: Quantitative Data Analysis, Statistical Modeling, Machine Learning/Computer Vision, Experiment Design & Benchmarking, Data Structures & Algorithms, Research & Literature Review, System Design, Test-Driven Development, Agile/Scrum